SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED
CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25
Question Count: 5

Code of Conduct
I Subash Surya N / 1924260164 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: Subashsurya N

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud.
   Discuss how cloud storage solutions ensure durability and data redundancy.
   How can encryption and access control protect sensitive backup data?
   Explain the cost advantages compared to traditional storage systems.
   Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage.
   Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment.
   Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments.
   Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

ASSESSMENT TOOL 3: Application Based Questions - Answers

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1. Auto-scaling and load balancing during seasonal sales

Scenario: An e-commerce site such as Amazon or Flipkart faces sudden traffic jumps during festival sales (for example Prime Day or Big Billion Days). Fixed servers would either crash at the peak or sit idle (and cost money) the rest of the year.

How auto-scaling and load balancing handle the demand

- **Auto-scaling** adds or removes virtual machines or containers automatically using rules. Target-tracking policies keep average CPU near a goal (say 60%); scheduled scaling adds capacity before a known sale start; predictive scaling uses past traffic patterns. When the sale ends, instances are removed so the company pays only for what it used.
- **Load balancing** (for example an Application Load Balancer) spreads incoming requests across all running instances in several availability zones, so no single server becomes a bottleneck. New instances are registered with the load balancer automatically.
- Together they give **horizontal elasticity**: more users means more servers behind one stable entry point.

Reducing downtime and improving user experience

- Health checks remove failed or slow instances from rotation, and traffic moves to healthy ones, so a server failure is invisible to shoppers.
- Spreading servers over multiple availability zones keeps the site alive even if a whole data center fails.
- Capacity grows before servers saturate, so pages load fast, carts are not lost and checkout does not time out. Faster pages mean fewer abandoned carts and higher sales.

Role of monitoring tools

Tools such as Amazon CloudWatch, Azure Monitor or Prometheus/Grafana collect metrics (CPU, memory, request count, latency, error rate), logs and traces. They **trigger the scaling policies**, raise alarms to the operations team when error rates or response times cross a threshold, give dashboards for the sale-day war room, and provide data to tune thresholds and control cost after the event.

Example flow: At 12:00 midnight the sale starts, requests jump 10x, CloudWatch sees CPU above 60%, auto-scaling launches more instances in minutes, the load balancer begins sending them traffic, and after the sale the extra instances are terminated.

2. Secure cloud backup storage

Durability and redundancy

Cloud object storage (Amazon S3, Azure Blob, Google Cloud Storage) stores each object as several copies or erasure-coded fragments across multiple disks and at least three availability zones. Checksums detect and repair corrupted data automatically, giving very high durability (S3 advertises 99.999999999%, "eleven nines"). Versioning protects against accidental deletion or overwriting, and cross-region replication keeps another full copy in a distant region.

Encryption and access control

- **Encryption in transit** with TLS, and **encryption at rest** with AES-256, using provider-managed keys or customer-managed keys in a key management service (KMS/HSM). Even if a disk or copy is stolen, the data is unreadable.
- **Access control** through IAM roles and policies following least privilege, multi-factor authentication, bucket policies and blocking of public access.

- **Object Lock / WORM** and MFA-delete make backups immutable so ransomware or an insider cannot erase them. Access logs (for example CloudTrail) give an audit trail.

Cost advantages over traditional storage

- No capital expenditure on tape libraries, SAN hardware, data-center space, power and cooling; the model is pay-as-you-go.
- No over-provisioning: capacity grows as data grows, and nothing is bought "just in case".
- Storage tiering moves old backups to cheap archive classes (Glacier, Azure Archive) automatically through lifecycle rules.
- No hardware refresh cycles, off-site tape transport or dedicated administrators for the storage hardware.

Disaster recovery use case

A hospital keeps nightly database backups in cloud storage replicated to a second region. When a fire or ransomware attack destroys the primary server room, the IT team restores the latest backup into virtual machines in the other region and brings the patient-record system back within hours (a low RTO) with at most a day of lost data (RPO), without having built a second physical data center.

3. CDNs and global deployment of a mobile app

How a CDN improves performance and reduces latency

- A CDN (CloudFront, Akamai, Cloudflare) keeps cached copies of content in **points of presence (PoPs)** around the world. A user is routed to the nearest PoP by DNS or anycast, so the round-trip distance, and therefore latency, is much smaller than going to a single origin server.
- Origin load drops because most requests are served from cache, and bandwidth cost falls.
- Optimizations such as compression, HTTP/2 or HTTP/3, persistent connections and TLS termination at the edge speed up delivery further.

Edge servers and user experience across regions

Edge servers serve images, app assets, API responses and video segments from nearby, giving faster start-up, less buffering and smoother interaction. They absorb traffic spikes and DDoS attacks before they reach the origin, and edge compute (Lambda@Edge, Cloudflare Workers) can run small functions such as authentication or personalization close to the user. Users in Asia, Europe and America all get similar speed.

Global availability from cloud providers

- Providers operate many **regions**, each with several availability zones, so the app can be deployed near each user group.
- Global traffic management (Route 53 latency or geo routing, global load balancers) sends users to the closest healthy region and fails over automatically.
- Multi-region databases and storage replication keep data available if a region goes down.

Example (streaming/gaming): A service like Netflix or Hotstar places popular videos on edge servers inside local ISPs, so playback starts quickly in HD without buffering during a cricket match. An online game such as PUBG or Fortnite delivers patches and assets through a CDN, and runs matchmaking and game servers in regional data centers, keeping ping low (ideally below 50 ms), which is critical for real-time play.

4. Compliance and data security for a financial institution

Regulatory compliance and auditing

- Providers follow the **shared responsibility model**: they secure the infrastructure (physical security, hypervisor, network), while the bank secures its data, identities and configuration.

- They hold independent certifications and attestations such as ISO 27001, SOC 1/2/3, PCI DSS, and support GDPR and local regulator guidelines (for example RBI in India). Compliance reports can be downloaded on demand (for example AWS Artifact) for auditors.
- Services such as AWS Config, Azure Policy and Security Hub continuously check resources against compliance rules and flag drift.

Role of logging, monitoring and encryption

- **Logging** (CloudTrail, Azure Activity Log) records who did what, when and from where, giving a tamper-evident audit trail for investigations and audits.
- **Monitoring** (CloudWatch, GuardDuty, SIEM tools) detects anomalies such as unusual logins or large data transfers and alerts in real time.
- **Encryption** of data at rest and in transit, with keys held in HSM-backed key management and rotated regularly, keeps financial data confidential and meets standards like PCI DSS.

Ensuring data sovereignty

- Choose cloud regions inside the required country or jurisdiction and store and process data only there.
- Use policies (for example service control policies or Azure Policy) to block creating resources or replicating data in other regions.
- Use customer-managed encryption keys stored in the same jurisdiction, classify and tag data, and check contracts and provider sovereign-cloud options.

Example: In an online fund transfer or card payment, the customer data travels over TLS, card numbers are tokenized, the transaction database is encrypted with HSM-protected keys, and every API call and administrator action is logged. A monitoring system flags a suspicious burst of transfers for fraud review, and all payment data is stored in an in-country region to meet RBI data-localization rules.

5. Containerization for application deployment

Containers vs virtual machines

Aspect	Virtual Machine	Container
Architecture	Each VM has a full guest OS on top of a hypervisor	Containers share the host OS kernel; only the app and its libraries are packaged
Size	Gigabytes	Megabytes
Start-up time	Minutes	Seconds or less
Performance	Hypervisor and guest OS add overhead	Near-native performance
Resource usage	High (dedicated CPU, RAM and OS per VM); fewer per host	Low; many more instances per host
Isolation	Strong, hardware-level	Process-level; weaker unless hardened

How orchestration (Kubernetes) simplifies deployment and scaling

- **Scheduling:** places containers on suitable nodes automatically based on CPU and memory needs.
- **Self-healing:** restarts failed containers and replaces unhealthy nodes.
- **Scaling:** the Horizontal Pod Autoscaler adds or removes replicas based on load; services provide built-in load balancing and service discovery.
- **Rolling updates and rollbacks:** new versions are released with zero downtime and reverted if they fail.
- **Declarative configuration:** the desired state is written in YAML files, so deployments are repeatable and fit CI/CD pipelines.

Portability across cloud environments

A container image bundles the code, runtime and dependencies, so it runs the same on a developer laptop, on-premises, AWS, Azure or Google Cloud, which removes "it works on my machine" problems. Managed Kubernetes services (EKS, AKS, GKE) share the same API, so workloads can move between clouds or run in hybrid and multi-cloud setups with little change, reducing vendor lock-in.

Example of a microservices application: An online food-delivery app is split into separate services: user/login, restaurant catalog, cart and order, payment, delivery tracking and notification. Each runs in its own container with its own database and can be developed, deployed and scaled independently; for example, the order service is scaled up at dinner time while the catalog service stays small. Kubernetes manages all of them and an API gateway routes client requests.